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SYDE 780 – Final Report

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Introduction

The COVID-19 pandemic represents one of the most significant global health crises in recent history, causing widespread disruption across global systems, societies, and healthcare infrastructure. In Ontario alone, COVID-19 resulted in ~1.65 million confirmed positive cases and more than 18 thousand deaths [1], placing significant strain on the Canadian healthcare system. It is estimated that post-COVID-19 concerns alone have cost the Canadian healthcare system $7.8 to $50.6 billion CAD [2]. Beyond its financial impact, COVID-19 also exacerbated existing health disparities which disproportionally affect regions of lower socioeconomic status [3]. Understanding how these disparities affect the outcome of COVID-19 may provide key insights into improving the management of Ontario healthcare resources and providing an equitable healthcare system to all.

Socioeconomic factors are an important determinant of COVID-19 outcomes. Investigating these factors can provide an idea of the resource needs of certain regions. Prior research has demonstrated that individuals in lower-income locations have greater chances of contracting COVID-19 and higher mortality rates/worse outcomes [3], [4]. These outcomes may be influenced by rural-urban differences. It is known that rural regions often have less access to quality healthcare, worse outcomes, and limited support or research to improve policies surrounding these services [5], [6]. Investigating the effect of income on COVID-19 incidence rates and how that may affect rural-urban regions can help influence future policies related to rural-urban healthcare decisions, enabling informed decisions.

This study employs a combination of generalized linear models, multivariable linear models, and generalized estimating equations to examine the relationship between income and COVID-19 incidence rates. Generalized and multivariable linear modeling approaches can address which income factors influence incidence rates most strongly. These investigations are supported by correlation and Wilcoxon tests to confirm the difference in values measured. The generalized estimating equations model will examine how income factors influence incidence rates over time. Additionally, these tests will also be done on subsets of the data that are divided into rural and urban regions. This will enable investigation into the differences between rural and urban regions regarding COVID-19 outcomes.

The objective of this study is to utilise datasets from the Confirmed Positive Cases of COVID-19 in Ontario and Public Health Ontario to examine income-related measures of incidence rates across rural and urban public health regions in Ontario. This study will specifically aim to (1) identify which income variables are strongly associated with COVID-19 outcomes, and (2) examine the difference in incidence rates between rural and urban regions.

Methods

Dataset Descriptions

The Confirmed Positive Cases of COVID-19 in Ontario dataset includes 1,708,846 data points and 28 variables. This data displays as row data with each row representing a COVID-19 case reported by a public health unit. The dataset includes information such as date of reported case, reporting public health unit, outcome of case, and demographic information (age and gender). This data is reported over a range of time from 2020 to 2024. Cases and outcomes of COVID-19 were also collected from Public Health Ontario COVID-19 data. This includes a range of data from 2020 to 2024 covering cases and deaths per week involving 7,107 data points and 10 variables. Additional variables were collected from the 2021 Census of Population in Canada for purposes of mapping income to public health unit regions. This data was collected from the health region subdivisions within the census data. Additionally, rural and urban regions were defined by the Rural Ontario Institute and used in this study [7].

Inclusion and Exclusion Criteria

Data visualization and analysis included data collected between the dates of January 1, 2020, to January 1, 2024. All other data outside of these dates were excluded. Additionally, data scraped from the census included the most recent data from 2020 and onwards. This reduced data points from 1,708,846 to 1,694,033. No data points were excluded from datasets taken from Public Health Ontario because the data was automatically collected between the inclusion dates. Additionally, around ~6% of data points were missing from Cases or episodes per 100k population in the Public Health Ontario cases dataset. These were not excluded because they represent surveillance weeks where there were no reported cases (instead of N/A).

Methods for Data Visualization and Analysis

In short, data visualization involved characterizing total cases, total deaths, cases per 100,000 population, and deaths per 100,000 population over age, gender, public health unit regions, and time. More information on data visualization can be found in the R markdown file provided as supplementary material. After data visualization, variables selected for analysis included public health units, cases, deaths, rural vs urban, and income. More information on the specific variables for data analysis are included in Table 1 provided below.

Data analysis first included an analysis of flattened total cases and deaths taken from the Confirmed Positive Cases of COVID-19 in Ontario dataset. Correlation tests on total cases and deaths were used as an initial identifier of potentially correlated variables. Linear models were used to display the correlated relationships between cases and deaths. This resulted in a $p < 0.005$ for total low-income measure and low-income measure after tax for cases and deaths, indicating that there is statistical evidence that a linear relationship exists between the variables. Linear plots were utilised to illustrate the relationship between the variables and adjusted for total population per public health unit. Additionally, multi-variable linear models were used to determine which factors most influenced cases and deaths. This resulted in a model reduction illustrating that prevalence of low-income measures after tax and total low-income measures had the most impact with $p < 0.005$ for cases. Low-income measures after tax, average household income after tax, and average total household income and p-values of $p < 0.005$, $p = 0.023$, and $p = 0.026$ for deaths.

Generalized linear regression was also used to analyse cases and deaths per 100,000 population against income variables. Model reduction was done to determine the most important income factors that affected the models. This was done on the data as a whole and the data divided into rural and urban. A Wilcoxon test was used on cases and deaths against rural and urban public health units to determine if there was a difference in the values for either classification of regions resulting in a p-value of $p < 0.005$. Finally, a generalized estimating equations model was used to analyze the influence of income on cases and deaths per 100,000 of the population.

Table 1. Definitions of variables included in analysis [1], [8], [9].

Term	Definition
Confirmed Positive Cases of COVID-19 in Ontario and Public Health Ontario Datasets	
Reporting_PHU/Public.health.unit	Public health unit associated with case(s) or death(s).
Rural_Urban	Indicates rural (0) or urban (1) public health unit.
Accurate_Episode_Date	An approximation of incidence onset date.
Age_Group	Age group of the patient.
Client_Gender	Gendier information of the patient.
Outcome1	Patient outcome of incidence.
Cases.or.episodes.per.100.000	Cases per 100,000 of the population per a 7-day period.
Number.per.100.000	Deaths per 100,000 of the population per a 7-day period.
2021 Census of Population Canada	
median_total	Total median individual income.
median_aftertax	Median individual income after tax.
average_total	Total average indiviudal income.
average_aftertax	Average individual income after tax.
median_total_household	Total median household income.
median_aftertax_household	Median household income after tax.
average_total_household	Total average household income.
average_aftertax_household	Average household income after tax.
total_LIM	Total low income measures. The income situation of the statistical unit in relation to a specific low-income line where units below the line are considered low income.
LIM_AT	Low income measures after tax. A fixed percentage (50%) of median-adjusted after-tax income of private households
prevalence_LIM_AT	Prevalence of low income measures after tax. The proportion or percentage of units whose income falls below a specified low-income line.

LIM = low-income measures, AT = after tax

Results

Population and Demographics

The data included in this study was collected from all 34 public health unit regions in Ontario. Data from the Confirmed Positive Cases of COVID-19 in Ontario included 1,720,242 observations (reduced to 1,694,033 from inclusion dates of January 1, 2020, to January 1, 2024) and data from Public Health Ontario included 7,108 observations over a period from 2020 to 2024. While the Public Health Ontario data did not provide demographic information beyond cases per 100,000 population, the Confirmed Positive Cases of COVID-19 in Ontario provided information on patient gender and age (Table 2). Additionally, cases were also characterized by public health unit region. No additional demographic information was provided.

Table 2. Demographic information from Confirmed Positive Cases of COVID-19 in Ontario.

Population	Female	Male	Gender Diverse	Unspecified	Total
Total	934503	748194	10	11326	1694033
<20s	117781	121255	1	2996	242033
20s	168186	133285	1	2547	304019
30s	152434	114088	3	1668	268193
40s	135716	94922	0	1311	231949
50s	124467	94448	0	1129	220044
60s	76856	71793	2	683	149334
70s	54339	54487	1	350	109177
80s	61417	44795	1	273	106486
90+	43143	18945	1	162	62251
Unknown	164	176	0	207	547

Income Variable Influences on COVID-19 Cases and Mortality

Income variables were investigated against COVID-19 cases and mortality for all 34 public health regions in Ontario. Initial correlation analysis revealed that individual income measures (especially median values) demonstrated weak correlation to incidence rates with values ranging from -0.097 to -0.073. Household income measures fared better with slightly higher positive correlation values to incidence rates ranging from 0.273 to 0.539. It is noted that median values demonstrated reduced correlation compared to their average counterparts. Conversely, low-income measures demonstrated high correlation (excluding prevalence of low-income measures). Total low-income measures and low-income measures after tax displayed correlation values of 0.990 and 0.971 for total cases. Correlation values for total deaths were 0.972 and 0.996. Additionally, total population of the public health unit regions was also strongly correlated with total cases and deaths (0.990 and 0.971, Table 3).

Table 3. Bivariate correlation test on income factors for total number of cases and deaths from Confirmed Positive Cases of COVID-19 dataset.

Variable	total_cases	total_deaths
median_total	-0.094	-0.097
median_aftertax	-0.073	-0.078
average_total	0.456	0.439
average_aftertax	0.397	0.383
median_total_household	0.390	0.273
median_aftertax_household	0.394	0.273
average_total_household	0.539	0.448
average_aftertax_household	0.521	0.418
total_LIM	0.990	0.971
LIM_AT	0.972	0.996
prevalence_LIM_AT	-0.099	-0.015
Total_Population	0.990	0.971

LIM = low-income measures, AT = after tax

Simple linear models were developed for the low-income measure variables with highest correlation to total cases and total deaths. These models confirmed that low-income measures are strongly associated with COVID-19 cases and deaths ($p < 0.001$). Essentially, these models suggest that regions of greater low-income measures are associated with higher incidence rates. An example of these relationships is demonstrated below with total low-income measure against percentage of incidence cases per population for each public health region (Table. 4).

Table 4. Simple linear models selected due to high correlation values with total number of cases and deaths. Analysis of linear relationship between low-income measures after tax and total low-income measures against total number of cases and deaths from the Confirmed Positive Cases of COVID-19 dataset.

Variables	Estimate	Std.Error	t value	Pr(>|t|)
total cases ~ low-income measures after tax				
(Intercept)	1317.44428	3687.78793	0.357	0.723
LIM_AT	1.14811	0.04929	23.295	p < 0.005
total deaths ~ low-income measures after tax				
(Intercept)	-58.063	16.883	-3.439	p < 0.005
LIM_AT	0.014	0.000	63.259	p < 0.005
total cases ~ total low-income measures				
(Intercept)	-8032.471	2371.659	-3.387	p < 0.005
total_LIM	0.139	0.004	38.995	p < 0.005
total deaths ~ total low-income measures				
(Intercept)	-143.517	48.285	-2.972	0.00557
total_LIM	0.002	0.000	22.783	p < 0.005

LIM = low-income measures, AT = after tax

Reduced multivariable linear models demonstrated that low-income measure variables were associated with increases in incidence rates for both cases and deaths (again). Total low-income measures for increased incidence cases and low-income measures after tax for increased incidence deaths had strong statistical evidence ($p < 0.001$) for association. Additionally, it seemed that higher income levels were associated with reduced incidence cases and deaths. For example, higher average individual income after tax was associated with a -1.745 decrease in incidence cases ($p = 0.005$). However, the relationship is complex as this model also demonstrates that higher average total household income is associated with reduced mortality rates ($p = 0.026$) while higher average household income after tax is associated with increased mortality rates ($p = 0.023$, Table 5).

Table 5. Reduced multivariable linear models on income variables as an indicator of total cases and deaths from the Confirmed Positive Cases of COVID-19 dataset.

Variables	Estimate	Std.Error	t value	Pr(>|t|)
total cases - multivariable linear model				
(Intercept)	51200.893	29076.420	1.761	p = 0.088
average_aftertax	-1.745	0.576	-3.029	p = 0.005
total_LIM	0.145	0.003	45.345	p < 0.001
prevalence_LIM_AT	1482.903	735.597	2.016	p = 0.053
total deaths - multivariable linear model				
(Intercept)	-512.200	200.211	-2.558	p = 0.016
average_total_household	-0.022	0.009	-2.337	p = 0.026
average_aftertax_household	0.031	0.013	2.389	p = 0.023
LIM_AT	0.014	0.000	56.796	p < 0.001

LIM = low-income measures, AT = after tax

Multivariable linear models were also created with data per 100,000 COVID-19 cases and deaths to see if there was a difference in the statistically significant income variables. This data was measured weekly over the period between 2020 to 2024. Again, higher income for individuals and households demonstrated a decrease in incidence cases (median total: $p = 0.005$, average total household: $p < 0.001$) but higher after-tax income demonstrated increased incidence cases (median after tax: $p = 0.003$, average after tax household: $p < 0.001$). A similar relationship was seen for average household income before and after tax for incidence deaths. This suggests even more evidence for a conflicting relationship between income and incidence rates. Low-income measures were associated with an increase in cases (prevalence of LIM-AT: $p < 0.001$) and a slight increase in incidence deaths (total LIM: $p = 0.002$, Table 6).

Table 6. Reduced multivariable linear models on income as an indicator of cases or deaths per 100,000 of the population from the Public Health Ontario COVID-19 cases and outcomes datasets.

Variables	Estimate	Std.Error	t value	Pr(>|t|)
cases per 100k - multivariable linear model				
(Intercept)	-221.919	36.748	-6.039	p < 0.001
median_total	-0.013	0.005	-2.839	p = 0.005
median_aftertax	0.017	0.006	2.956	p = 0.003
average_total_household	-0.003	0.001	-3.77	p < 0.001
average_aftertax_household	0.005	0.001	4.11	p < 0.001
prevalence_LIM_AT	5.588	0.874	6.397	p < 0.001
deaths per 100k - multivariable linear model				
(Intercept)	0.383	0.222	1.729	p = 0.084
median_aftertax_household	-2.44E-05	8.57E-06	-2.847	p = 0.004
average_total_household	-2.70E-05	9.77E-06	-2.768	p = 0.006
average_aftertax_household	5.45E-05	1.94E-05	2.804	p = 0.005
total_LIM	1.07E-07	3.41E-08	3.132	p = 0.002

LIM = low-income measures, AT = after tax

Rural and Urban Influences on COVID-19 Cases and Mortality

Rural and urban influences were investigated by categorizing the public health regions into their respective classification based off the Ontario Rural Institute definition of urban and rurality. This resulted in 15 rural public health regions and 19 urban public health regions. Generalized linear models were created for rural and urban regions, comparing cases and deaths per 100,000 of the population.

The rural regions displayed more complex associations with higher income often associated with reduced incidence cases (median total: $p = 0.007$, average total household: $p < 0.001$) and higher income after tax associated with increased incidence cases (median after tax: $p = 0.037$, average after tax household: $p < 0.001$). Additionally, higher average total individual income was also associated with increased incidence cases ($p < 0.001$). Low-income measure variables also displayed a contradicting trend with total low-income measures increasing incidence cases ($p = 0.025$) and low-income measures after tax decreasing incidence cases ($p = 0.010$). Prevalence of low-income measures after tax had strong evidence that it was associated with increased incidence rates ($p < 0.001$, Table 7).

On the other hand, urban regions did not display any association with low-income measure variables. However, these regions also displayed the trend of high-income reducing incidence cases (median total: $p < 0.001$, average total: $p = 0.008$) while high-income after-tax increased incidence rates (average after tax: $p = 0.005$, Table 7).

Table 7. Reduced multivariable linear models on income as an indicator of cases per 100,000 of the population between rural and urban regions from the Public Health Ontario COVID-19 cases and outcomes datasets.

Variables	Estimate	Std.Error	t value	Pr(>|t|)
cases per 100k (rural) - multivariable linear model				
(Intercept)	-663.218	127.557	-5.199	p < 0.001
median_total	-0.027	0.010	-2.696	p = 0.007
median_aftertax	0.023	0.011	2.088	p = 0.037
average_total	0.014	0.004	3.393	p < 0.001
average_total_household	-0.027	0.008	-3.508	p < 0.001
average_aftertax_household	0.034	0.010	3.564	p < 0.001
total_LIM	0.001	0.000	2.248	p = 0.025
LIM_AT	-0.006	0.002	-2.592	p = 0.010
prevalence_LIM_AT	9.088	2.535	3.585	p < 0.001
cases per 100k (urban) - multivariable linear model				
(Intercept)	-22.904	48.483	-0.472	p = 0.637
median_total	-0.004	0.001	-3.888	p < 0.001
average_total	-0.008	0.003	-2.639	p = 0.008
average_aftertax	0.015	0.005	2.819	p = 0.005

LIM = low-income measures, AT = after tax

When comparing incidence mortality rates per 100,000 of the population, rural regions also displayed complex relationships. For example, higher average total individual income ($p < 0.001$) and median total household income ($p = 0.004$) were associated with increased mortality rates. Average individual income after tax ($p < 0.001$) and median household income after tax ($p = 0.008$) were associated with reduced mortality rates. Conversely, average total household income ($p < 0.001$) and average household income after tax ($p < 0.001$) flipped this narrative. Additionally, there were no low-income measure variables that were statistically significant (Table 8).

In urban regions, higher average total household income ($p < 0.001$) and median total household income ($p < 0.001$) were associated with reduced mortality rates. Consequently, average individual income after tax ($p < 0.001$) was associated with an increase in mortality rates. Again, higher average total household income broke this trend by associating with increased mortality rates too ($p < 0.001$). Low-income measures after tax played a role in associating with increased mortality rates ($p = 0.010$, Table 8).

Table 8. Reduced multivariable linear models on income as an indicator of deaths per 100,000 of the population between rural and urban regions from the Public Health Ontario COVID-19 cases and outcomes datasets.

Variables	Estimate	Std.Error	t value	Pr(>|t|)
deaths per 100k (rural) - multivariable linear model				
(Intercept)	1.018	0.971	1.049	p = 0.294
average_total	0.002	0.001	3.876	p < 0.001
average_aftertax	-0.003	0.001	-3.834	p < 0.001
median_total_household	1.54E-04	5.26E-05	2.922	p = 0.004
median_aftertax_household	-2.05E-04	7.78E-05	-2.641	p = 0.008
average_total_household	-0.001	3.46E-04	-3.999	p < 0.001
average_aftertax_household	0.002	4.16E-04	3.948	p < 0.001
deaths per 100k (urban) - multivariable linear model				
(Intercept)	-0.406	0.513	-0.791	p = 0.429
average_total	-1.43E-04	3.89E-05	-3.685	p < 0.001
average_aftertax	1.77E-04	5.22E-05	3.396	p < 0.001
median_total_household	-2.86E-05	8.29E-06	-3.447	p < 0.001
average_total_household	2.90E-05	8.67E-06	3.35	p < 0.001
LIM_AT	8.60E-07	3.35E-07	2.565	p = 0.010

LIM = low-income measures, AT = after tax

Finally, a Wilcoxon test revealed that there is a statistically significant difference in both cases and mortality incidence rates between rural and urban regions. On average, urban regions had slightly higher incidence case and mortality rates in comparison to rural regions ($p = 0.002$ and $p < 0.001$, respectively, Table 9).

Table 9. Wilcoxon test on average number of cases and deaths per 100,000 of the population between rural and urban regions from the Public Health Ontario COVID-19 cases and outcomes datasets.

Variables	Rural	Urban	U-test
Cases per 100k population	54.708	55.178	p = 0.002
Deaths per 100k population	0.530	0.538	p < 0.001

Analysis Over Time

A generalized estimating equations model was developed to investigate changes in cases and mortality rates over time. This comparison was done on the whole dataset in addition to rural and urban datasets. While the majority of the models displayed poor residual plots with clearly defined linear patterns (not the random scatter expected for good models), one model, in particular, showed a particularly strong relationship. The rural dataset for incidence cases demonstrated a significantly high robust Z value for low-income measure variables with statistically significant evidence ($p < 0.001$). This highlights the importance of low-income measure variables on COVID-19 incidence rates (Table 10).

Table 10. Generalized estimating equations model on analysis of income variables as an indicator of cases per 100,000 of the population in rural regions. This data was analyzed over a range from 2020 to 2024 with specific snapshots taken at surveillance weeks of 10, 20, 30, 40, and 50.

Term	Estimate	Naive_SE	Naive_z	Robust_SE	Robust_z	Pr(>|t|)
(Intercept)	-681.795	90.537	-7.531	46.910	-14.534	p < 0.001
Surveillance.week	-0.263	0.246	-1.071	0.286	-0.922	p = 0.357
median_total	0.001	0.008	0.074	0.005	0.118	p = 0.906
median_aftertax	-0.001	0.008	-0.127	0.005	-0.199	p = 0.842
average_total	-0.035	0.041	-0.860	0.032	-1.103	p = 0.270
average_aftertax	0.050	0.047	1.056	0.037	1.337	p = 0.181
median_total_household	-0.002	0.003	-0.674	0.002	-1.149	p = 0.250
median_aftertax_household	-0.004	0.005	-0.828	0.004	-1.185	p = 0.236
average_total_household	-0.001	0.022	-0.066	0.017	-0.087	p = 0.931
average_aftertax_household	0.010	0.025	0.407	0.019	0.534	p = 0.593
total_LIM	0.001	0.000	4.767	0.000	8.955	p < 0.001
LIM_AT	-0.010	0.002	-5.557	0.001	-11.238	p < 0.001
prevalence_LIM_AT	13.737	1.842	7.455	0.847	16.211	p < 0.001

LIM = low-income measures, AT = after tax

Discussion

This investigation was primarily designed to investigate income as a factor associated with the outcome of COVID-19 incidence rates. It was determined that low-income measures were consistently strong predictors of COVID-19 cases and mortalities across the different tests performed. Additionally, this study investigated the differences in incidence rates displayed by rural and urban public health regions. Urban regions showed a general trend of higher income reducing cases and mortality while rural regions demonstrated many contradicting results.

Initial tests determined that low-income measures were highly correlated with incidence cases and mortality rates. This was also confirmed through the final test using generalized estimating equations models. Regions of higher socioeconomic disparity were associated with higher incidence rates which agrees with previous literature that has also provided similar evidence [3]. Furthermore, this study captured a contradicting trend suggesting that regions with higher income had reduced incidence rates while regions with higher income after tax had increased incidence rates. This may be the result of progressive tax policies aimed at low-income regions. However, previous research has also come to various conflicting conclusions on the outcomes of COVID-19 incidences.

Little et al. (2021) measured impact of neighborhood poverty levels on COVID-19 outcomes and found that residents of high-poverty regions (>20% of the population lived below the federal poverty threshold line) were not linked to higher rates of COVID-19 hospitalizations and had reduced rates of in-hospital mortality [3]. Conversely, Abedi et al. (2021) determined that regions with higher percentage of the population living under the poverty level were highly associated with greater mortality whereas regions of greater socioeconomic status had greater rates of cases but reduced mortality [4]. Ingen et al. (2022) found that marginalized communities, such as those with visible minorities, had 4 times the higher risk of cases while Buja et al (2020) argued that incidence case rates were associated with factors such as population density and levels of employment [10], [11].

This study found evidence supporting the difference in COVID-19 incidence outcomes between rural and urban regions. Essentially, urban income trends seemed to follow the same pattern suggesting that higher income reduced incidence rates while higher income after tax increased rates. Rural income trends were more scattered with no clear pattern followed. Additionally, an overall comparison of COVID-19 outcomes between rural and urban regions determined that urban regions tended to have higher incidence rates for cases and deaths. These findings are generally supported by previous research in terms of cases. Cuadros et al. (2021) suggested that incidence rates are linked to time, demonstrating that incidence rates slowly increased in intensity between COVID waves for rural regions in America [12]. Additionally, Wachtler et al. (2020) agreed with this sentiment, suggesting that populations with less socioeconomic status were often harder hit harder in later stages of the pandemic [13]. However, this study did not fully align with previous research stating that regions of greater economic wealth, often urban areas, had reduced mortality rates as mentioned by Abedi et al. (2021). This may be related to how rural and urban regions were defined in this study as prior research may have defined these locations in greater detail rather than looking at them by public health region.

The limitations of this study include the generalizations and assumptions made due to the limited, non-specific data provided. The only demographic indicators attached to the data were age, gender, and location based on public health region. Data was not collected at an individual level meaning that all other factors (race, education, etc.) were generalized to the health region level from the 2021 Canadian census data. However, this dataset is complete with minimal missing or invalid values which lends to the results credibility. This study evaluated income which produced some conflicting trends in terms of incidence rates. Broadening the models used in this study by including additional factors may clarify the results. Finally, this study only included the 34 public health regions of Ontario. Further work should seek to find datasets that break down those regions into smaller subsets or expand to other regions in Canada. Increasing the specificity or breadth of regions examined will strengthen the rural-urban statistical results.

Conclusion

To conclude, this study demonstrated clear evidence that low-income measure variables have a high association with increased COVID-19 incidence rates in Ontario. Additionally, there is a difference in rural and urban outcomes of COVID-19 cases and mortality rates with income inequality playing a factor. Although this study provided evidence that urban regions experienced higher COVID-19 incidence rates, addressing rural-urban disparities and income inequality are critical steps toward improving the Canadian healthcare system.

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